

Circuit objectives

Thursday, August 20, 2026 8:59 PM

Overview

The core challenge of the circuit is this: **the photodiodes produce an incredibly tiny current, buried in a sea of infrared noise from the sun, lights, and the environment.** The entire analog signal chain exists to solve that one problem — extract *your* signal from all the noise, and turn it into something the STM32 can act on.

The Fundamental Principle — Why 38 kHz?

Nothing in nature pulses infrared light at exactly 38 kHz. Sunlight is constant DC. Indoor lighting flickers at 50/100 Hz. By modulating the IR LED at **38 kHz**, the system creates a unique "fingerprint" that only your light has. Every circuit stage downstream is then tuned to listen *exclusively* for that fingerprint and reject everything else.

The Signal Chain — Stage by Stage

Stage 0 — IR LED Driver

- An NPN transistor (2N3904) switches the IR LED (TSAL6400) on and off at **38 kHz**, driven by a PWM signal from the STM32
- The LED blasts invisible infrared light outward continuously while the system is active
- A current-limiting resistor sets the LED drive current to a safe operating level
- When the light hits the retroreflective sticker on the target, it bounces directly back toward the turret

Stage 1 — Photodiode (The Sensor)

- Four SFH203 photodiodes are arranged in a **cross pattern** — one facing top, bottom, left, and right
- When infrared light hits a photodiode, it generates a tiny electrical current. More light = more current
- The key insight is **differential comparison**: when the current from the left photodiode equals the current from the right, and top equals bottom, the turret is perfectly centred on the target. The system is constantly computing this difference
- At this stage the signal is just a **microampere-level current** — far too small and too noisy to do anything useful with directly

Stage 2 — Transimpedance Amplifier (TIA)

- **Problem:** The photodiode output is a current, not a voltage. Microcontrollers read voltages, not currents. The signal is also extremely weak.
- **Solution:** The TIA (built around an OPA2134 op-amp with a feedback resistor) does two things simultaneously:
 - **Converts** the photodiode current into a voltage: $V_{out} = I_{photodiode} \times R_f$
 $V_{out} = I_{photodiode} \times R_f$
 - **Amplifies** that voltage to a measurable level
- Each of the four photodiodes has its own dedicated TIA circuit
- At this point the signal is a proper voltage, but it still contains **all frequencies** — your 38 kHz signal, sunlight, room lighting, everything mixed together

Stage 3 — Active Bandpass Filter (Sallen-Key Topology)

- **Problem:** The amplified voltage contains noise from every IR source in the environment — sunlight alone would saturate the signal
- **Solution:** A Sallen-Key active bandpass filter is tuned precisely to **38 kHz** with a defined Q factor. It passes only signals at that frequency and aggressively attenuates everything else
- Sunlight (DC), indoor lighting (50/100 Hz), and all other interference are rejected
- What comes out the other side is a **clean 38 kHz sine wave** — and only that —

whose amplitude corresponds to how much of *your* IR light is hitting that specific photodiode

- Each photodiode channel has its own bandpass filter
- This is the circuit that makes the system work in daylight

Stage 4 — Comparator with Hysteresis

- **Problem:** The filtered signal is still an analog sine wave. The STM32 needs a clean digital answer: *is there a target in front of this photodiode right now, yes or no?* Additionally, if the signal hovers right at the detection threshold, the output would flicker rapidly between yes and no — causing the motors to jitter uncontrollably
- **Solution:** An LM393 comparator with a **hysteresis resistor** (positive feedback) solves both problems:
 - It compares the filtered signal against a reference threshold voltage
 - It outputs a clean digital **HIGH** (target detected) or **LOW** (no target)
 - The hysteresis creates a *deadband* — the signal must rise above the upper threshold to trigger HIGH, and must fall below a lower threshold to return to LOW. This prevents flickering at the boundary
- Each photodiode channel has its own comparator — four comparators total
- The comparator stage also acts as a **noise gate**: if there is no target at all, no 38 kHz signal gets through the bandpass filter, the comparator stays LOW, and the STM32 receives four LOW signals — the motors do not move. This prevents the turret from thrashing around randomly when no target is present

Stage 5 — Differential Signal Processing

- The four comparator outputs (TOP, BOTTOM, LEFT, RIGHT) feed into the STM32 ADC
- The STM32 also reads the raw analog amplitudes from the TIA outputs to compute a **normalised bearing error**:

$$e_{pan} = I_{left} - I_{right} \quad I_{left} + I_{right} \quad e_{pan} = I_{left} + I_{right} \quad I_{left} - I_{right}$$

$$e_{tilt} = I_{top} - I_{bottom} \quad I_{top} + I_{bottom} \quad e_{tilt} = I_{top} + I_{bottom} \quad I_{top} - I_{bottom}$$

- Normalising by the total signal strength means the error signal is **independent of how far away the target is** — the system responds to *direction*, not distance
- These error values feed directly into the PID controller to drive the servo motors

Full Signal Chain Flow

IR LED pulses at 38 kHz

↓

Light strikes retroreflective marker on target

↓

Reflected IR returns to the four photodiodes

↓

Each photodiode generates a tiny current (μA range)

↓

TIA converts current → voltage and amplifies it

↓

Bandpass filter (38 kHz, Sallen-Key) strips out all ambient IR noise — only your signal passes through

↓

Comparator + hysteresis converts to clean HIGH/LOW
— noise gate prevents motor jitter with no target
— hysteresis prevents flickering at the threshold

↓

STM32 reads all 4 signals and computes bearing error

↓

Left stronger than right? → pan error → rotate left
Top stronger than bottom? → tilt error → tilt up

↓

PID controller computes servo commands

↓

Servo motors correct turret position

↓

Loop repeats 1,000 times per second

IMU Circuit — Body Motion Compensation

The Problem the IMU Solves

When you walk or turn, your shoulder moves — and the turret moves with it. Without compensation, every step you take would throw the turret off target. The IMU detects your body movement so the system can **pre-correct** for it before the photodiodes even need to react.

What the IMU Contains

The MPU-9250 is a single IC containing three sensors:

- **Gyroscope** — measures angular velocity (how fast you are rotating) in X, Y, Z. It tells you *speed of rotation right now*, not absolute position. To get angle you must integrate: $\theta(t) = \int 0t \omega(\tau) d\tau$ $\theta(t) = \int 0t \omega(\tau) d\tau$. The problem is this integration accumulates error over time — this is called **gyroscope drift**
- **Accelerometer** — measures forces including gravity. When stationary, it tells you which way is down, giving an absolute orientation reference. It is accurate long-term but noisy during fast movement
- **Magnetometer** — measures the Earth's magnetic field for absolute heading reference

Why You Cannot Use Either Sensor Alone

Sensor	Good At	Bad At
Gyroscope	Fast, short-term rotation	Drifts over time
Accelerometer	Long-term, stationary reference	Noisy during fast movement

They are complementary. Neither is sufficient alone.

The Madgwick Filter — Fusing Both Sensors

The **Madgwick filter** is the algorithm that combines gyroscope and accelerometer data intelligently to get the best of both:

- **Trust the gyroscope** for fast, short-term rotation tracking
- **Trust the accelerometer** to slowly correct the gyroscope's drift over time
- The output is a clean, stable **orientation quaternion** q representing the true shoulder orientation, updated via:

$$\dot{q} = \frac{1}{2} q \otimes \omega q^* = \frac{1}{2} q \otimes \omega$$

where ω is the angular velocity vector from the gyroscope, corrected by the accelerometer feedback term

How It Interfaces to the Circuit

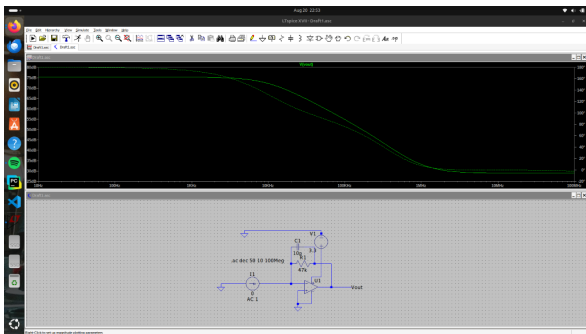
- The MPU-9250 communicates with the STM32 via **SPI** — the STM32 sends a read request and the IMU returns six numbers: acceleration in X, Y, Z and angular velocity in X, Y, Z
- This happens **200 times per second**
- A proper **decoupling network** (100 nF + 10 μ F capacitors, ferrite bead on the power supply line) is placed on the PCB directly at the IMU power pins — the IMU is sensitive to power supply noise and without this the readings will be corrupted
- The clean orientation data is passed to the tracking controller as a **feedforward term**: if the shoulder rotates 10° left, the turret is immediately commanded 10° right to compensate — keeping it approximately on target *during* the movement, after which the photodiodes fine-tune the lock

Transimpedance amplifier

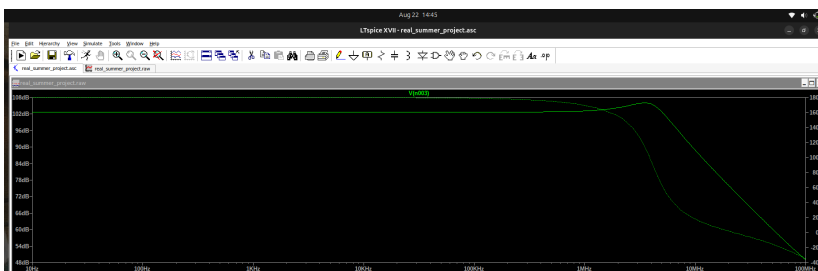
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In real life op amps are not ideal which is why TIAs are frequency dependent

This gave the correct graph - TIAs are inherently low pass filters. My blinks will be at 38kHz so I want to make the gain drop off after 38kHz. This is entirely dependent on the op amp, not the surrounding circuitry. Need to make drop off start after 38kHz.

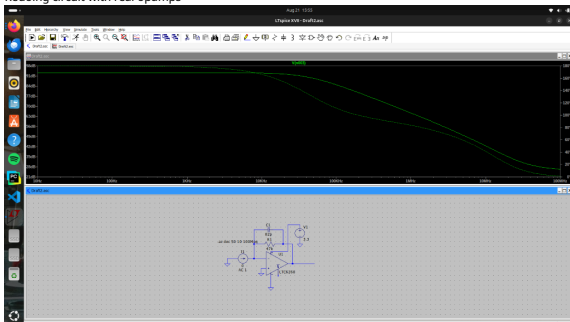


After changing the



Bump after - need to add capacitor to remove it. I will add a capacitor to improve stability, reduce high frequency gain and reduce rms noise

Redoing circuit with real opamps



Now gain starts going down after 38kHz

What A TIA Actually Does

A TIA has **one job**:

$$V_{out} = I_{in} \times R_f \quad V_{out} = I_{in} \times R_f$$

That's it. The op-amp's role is to **force its inverting input to 0V** (virtual ground), so all the photodiode current is forced through $R_f R_f$, creating a voltage. There is no voltage input — so talking about "voltage gain" is meaningless here.

Why Your Numbers Are Fine

Your circuit has:

- $R_f = 100 \text{ k}\Omega$ $k \Omega R_f = 100 \text{ k}\Omega$
- $I_{PD, max} \approx 10 \text{ }\mu\text{A}$ $AIPD, max \approx 10 \text{ }\mu\text{A}$ (closest realistic range)

So maximum output:

$$V_{out, max} = 10 \text{ }\mu\text{A} \times 100 \text{ k}\Omega = 1 \text{ V} \quad V_{out, max} = 10 \text{ }\mu\text{A} \times 100 \text{ k}\Omega = 1 \text{ V}$$

That's **well within** your supply rail. ✓

What The 100 dB Ω Actually Means

When LTspice runs an AC analysis with a current source input, it computes:

$$\text{Plot} = 20 \log_{10}(V_{out}/I_{in}) = 20 \log_{10}(R_f) \quad \text{Plot} = 20 \log_{10}(I_{in}/V_{out}) = 20 \log_{10}(R_f) \\ = 20 \log_{10}(100 \times 10^3) = 100 \text{ dB}\Omega = 20 \log_{10}(100 \times 10^3) = 100 \text{ dB}\Omega$$

This is **not voltage gain**. It's just $R_f R_f$ expressed in dB. The Ω unit is attached — it's a completely different quantity to V/V.

The Analogy

Think of it this way:

Amplifier Type	Input	Output	Gain Units
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Voltage amplifier	Voltage	Voltage	V/V (dB)
TIA	Current	Voltage	Ω (dB Ω)

Saying your TIA has "100dB voltage gain" is like saying a resistor has "100dB voltage gain" just because $V = IR$. It doesn't — it just converts current to voltage via Ohm's law. **Your TIA is doing exactly the same thing**, just with an op-amp to provide buffering and virtual ground.

Why Your Plot Confirms It's Working

Feature On Your Plot	What It Confirms
Flat at ~96–100 dB Ω	$Rf = 100k\ \Omega$ $Rf \approx 100k\Omega$ is dominating as expected ✓
Flat all the way through 38kHz	Your signal passes with no attenuation ✓
Rolloff starts in MHz range	$Cf = 1pF$ $Cf \approx 1pF$ limiting bandwidth correctly ✓
No peaking before rolloff	Circuit is stable, not oscillating ✓

Bottom Line

Your TIA is correct because:

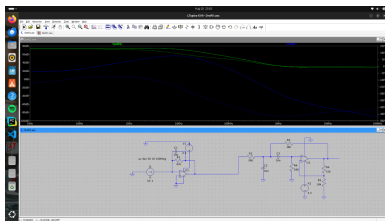
- $V_{out} = IPD \times Rf$ $V_{out} = IPD \times Rf$ gives **safe output voltages** at all operating distances
- The **bandwidth is flat at 38kHz** — your signal passes through unattenuated
- The circuit is **stable** — no peaking or ringing visible
- The **100 dB Ω** is just Rf in dB — it is not a dangerous voltage gain

Ready to move to the bandpass filter?

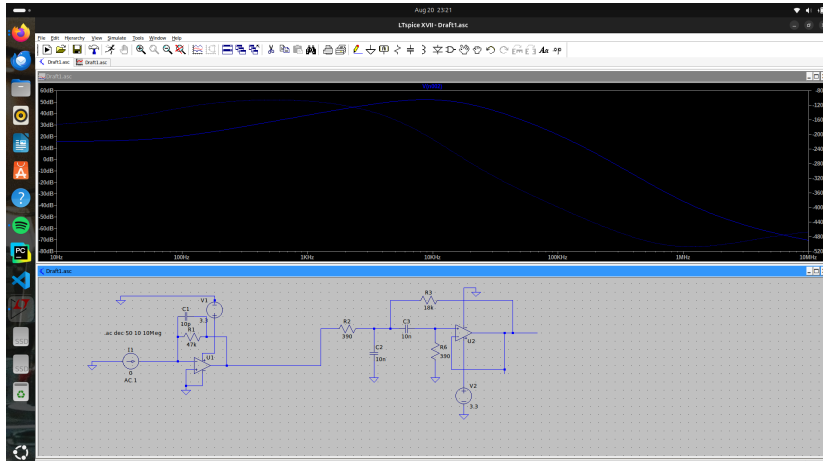
.acBandpass filter

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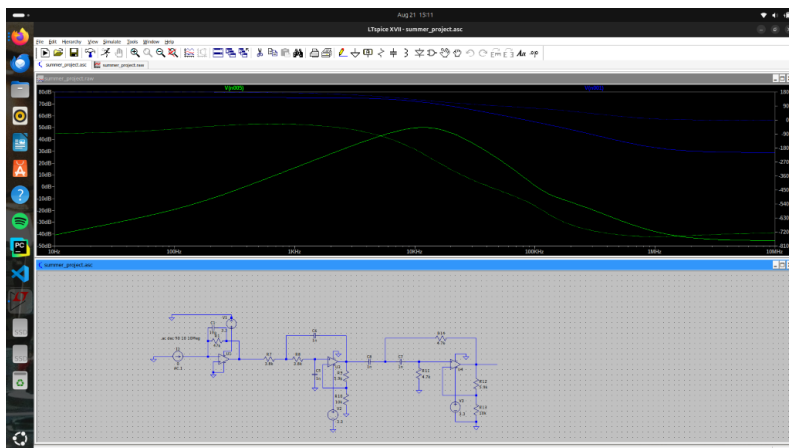
Since we are interested in frequencies at 38kHz, we are going to try filter out everything else.



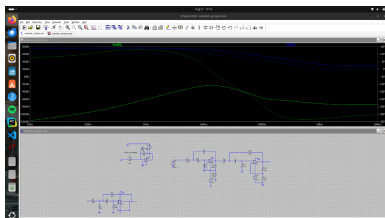
This is peaking at around 10kHz. We want it to peak around 38kHz.



Cant pinpoint the problem so im gonna redesign my own circuit from scratch.



This is making me think that because the TIA drops off before 38kHz, that's causing the issue, so im going to remove the TIA and test this circuit itself.



Same issue has occurred

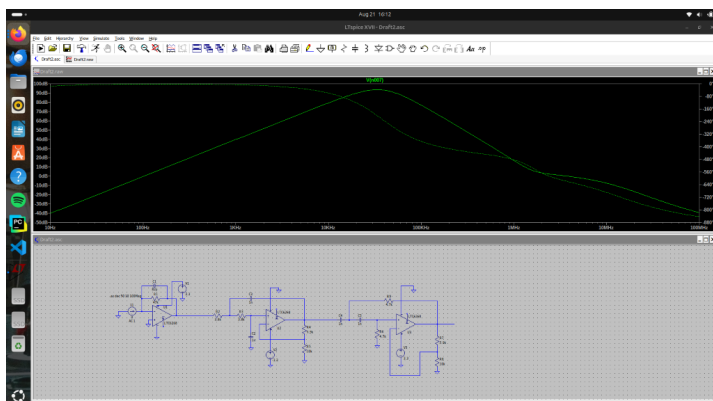
I noticed that in the TIA where both the drop started and also where the peak was in the bandpass filter was always the same, independent of resistor values. Even when trying different configurations of bandpass filters so I decided to change the opamp. This is what fixed the issue.

The simulated active bandpass filter peaked at $\approx 10\text{ kHz}$ instead of the designed 40 kHz when using UniversalOpamp2, but aligned with theoretical calculations upon substituting a manufacturer op-amp model.

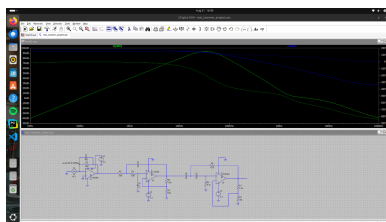
Root Cause Analysis

- **Gain-Bandwidth Limitation:** UniversalOpamp2 defaults to a fixed Gain-Bandwidth Product ($\text{GBW} = 10\text{ MHz}$). In active filter topologies, insufficient op-amp bandwidth introduces excess open-loop phase lag into the feedback network, which shifts closed-loop pole positions and pulls the center frequency (f_0) significantly downward.
- **Internal Pole / Phase Margin:** The default phase margin setting (45°) adds a secondary high-frequency pole that exacerbates phase shift and dampens the intended 40 kHz resonance.

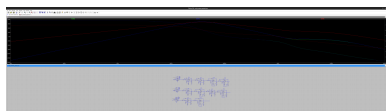
- **Component Discrepancy:** The selected physical op-amp model had a higher effective GBW and distinct phase compensation capable of driving the feedback network at 40 kHz without severe pole-shifting.



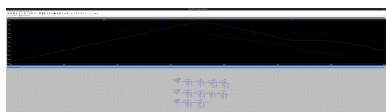
Now the peak is at 38kHz, as wanted, however the gain is far too high, and the roll off is too shallow for real world applications.



The blue line is the gain after the TIA, im going to try lower it there and it should lower the overall gain.



Cascading bandpass filters reduces the slope 0 couldve made a 4th order filter but this is more time efficient and achieves the same result. I tried different configurations of the circuit, low low high high and low high low high, both produce identical results.



When comparing the 4th order vs second order filter, we can see that the slope is far steeper

Filter	Attenuation at 10Hz
Single stage (blue)	-30 dB
Cascaded (green)	-170 dB

The difference:
 $-170 - (-30) = 140\text{ dB}$ extra rejection
 In linear terms that's:
 $10^{(140/20)} = 31,622,776$ better rejection

In nature, optical infrared (IR) noise is overwhelmingly **far less than 38 kHz**—it is almost entirely **DC (0 Hz) up to a few tens of Hertz**.

This means our filter will work amazingly

Choosing between 100k and 200k for the TIA

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Step 1 — How Much Current Does The Photodiode Produce?

When IR light hits the photodiode it produces a current. The formula is:

$$IPD = R_e \times E_e \times A \quad IPD = R_e \times E_e \times A$$

Where:

- $R_e = 0.5 \text{ A/W}$ $R_e = 0.5 \text{ A/W}$ — responsivity of SFH203 at 950nm
- E_e — irradiance (W/m^2) hitting the photodiode
- $A = 7.5 \times 10^{-6} \text{ m}^2$ $A = 7.5 \times 10^{-6} \text{ m}^2$ — active area of SFH203

► Finding The Irradiance From Your TSAL6400 LED

$$E_e = I_e / d^2 \quad E_e = I_e / d^2$$

Where:

- $I_e = 35 \text{ mW/sr}$ $I_e = 35 \text{ mW/sr}$ — radiant intensity of TSAL6400 at 100mA
- d = distance to target

So at different distances:

$$E_e(0.5\text{m}) = 0.0350 / 0.5^2 = 0.14 \text{ W/m}^2 \quad E_e(0.5\text{m}) = 0.0350 / 0.5^2 = 0.14 \text{ W/m}^2$$

$$E_e(0.1\text{m}) = 0.0350 / 0.1^2 = 3.5 \text{ W/m}^2 \quad E_e(0.1\text{m}) = 0.0350 / 0.1^2 = 3.5 \text{ W/m}^2$$

Therefore The Photodiode Current Is:

$$IPD(0.5\text{m}) = 0.5 \times 0.14 \times 7.5 \times 10^{-6} = 0.53 \text{ } \mu\text{A} \quad IPD(0.5\text{m}) = 0.5 \times 0.14 \times 7.5 \times 10^{-6} = 0.53 \text{ } \mu\text{A}$$

$$(0.5\text{m}) = 0.5 \times 0.14 \times 7.5 \times 10^{-6} = 0.53 \text{ } \mu\text{A}$$

$$IPD(0.1\text{m}) = 0.5 \times 3.5 \times 7.5 \times 10^{-6} = 13 \text{ } \mu\text{A} \quad IPD(0.1\text{m}) = 0.5 \times 3.5 \times 7.5 \times 10^{-6} = 13 \text{ } \mu\text{A}$$

So your photodiode produces currents in the range of **hundreds of nanoamps to low microamps** — tiny signals that need amplifying.

Step 2 — What Does The TIA Do With That Current?

The TIA converts that tiny current into a voltage using one simple relationship:

$$VTIA = IPD \times R_f \quad VTIA = IPD \times R_f$$

With $R_f = 100 \text{ k}\Omega$ $R_f = 100 \text{ k}\Omega$:

$$VTIA(0.5\text{m}) = 0.53 \text{ } \mu\text{A} \times 100 \text{ k}\Omega = 53 \text{ mV} \quad VTIA(0.5\text{m}) = 0.53 \text{ } \mu\text{A} \times 100 \text{ k}\Omega = 53 \text{ mV}$$

$$VTIA(0.1\text{m}) = 13 \text{ } \mu\text{A} \times 100 \text{ k}\Omega = 1.3 \text{ V} \quad VTIA(0.1\text{m}) = 13 \text{ } \mu\text{A} \times 100 \text{ k}\Omega = 1.3 \text{ V}$$

Why Is 100k Ω Suitable?

Two reasons:

1 — No clipping at close range: $VTIA, \text{max} = 13 \text{ } \mu\text{A} \times 100 \text{ k}\Omega = 1.3 \text{ V}$ $VTIA, \text{max} = 13 \text{ } \mu\text{A} \times 100 \text{ k}\Omega = 1.3 \text{ V}$ $1.3 \text{ V} < 3.3 \text{ V supply}$ $1.3 \text{ V} < 3.3 \text{ V supply}$ ✓

2 — Signal is strong enough at long range: $VTIA(0.5\text{m}) = 53 \text{ mV}$ $VTIA(0.5\text{m}) = 53 \text{ mV}$ This is detectable — not too small for the bandpass filter to work with ✓

What Does 100 dB Ω Mean?

This is just R_f expressed in dB:

$$Z_T = 20 \log_{10}(R_f) = 20 \log_{10}(100 \times 10^3) = 100 \text{ dB} \quad Z_T = 20 \log_{10}(R_f) = 20 \log_{10}(100 \times 10^3) = 100 \text{ dB}$$

It is **not a voltage gain** — it just means for every **1A of input current** you get **100,000V out**. Since your currents are in microamps, your output voltages stay safely small.

Step 3 — What Does The Bandpass Filter Add?

The Sallen-Key bandpass filter does two things:

1. **Rejects everything except 38kHz** — kills sunlight, fluorescent noise etc

2. Adds voltage gain at 38kHz — boosts your signal further

The filter has its own voltage gain A_v at 38kHz. Say your filter has a gain of **10 V/V** at the peak:

$$20 \log_{10}(10) = 20 \text{ dB} \quad 20 \log_{10}(10) = 20 \text{ dB}$$

So the **total system gain** after TIA + filter is:

$$G_{\text{total}} = G_{\text{TIA}} + G_{\text{filter}} = 100 \text{ dB} + 20 \text{ dB} = 120 \text{ dB} \quad G_{\text{total}} = G_{\text{TIA}} + G_{\text{filter}} = 100 \text{ dB} + 20 \text{ dB} = 120 \text{ dB}$$

And the output voltage after the filter:

$$V_{\text{filter}}(0.5\text{m}) = 53 \text{ mV} \times 10 = 530 \text{ mV} \quad V_{\text{filter}}(0.5\text{m}) = 53 \text{ mV} \times 10 = 530 \text{ mV}$$

$$V_{\text{filter}}(0.1\text{m}) = 1.3 \text{ V} \times 10 = 13 \text{ V} \quad V_{\text{filter}}(0.1\text{m}) = 1.3 \text{ V} \times 10 = 13 \text{ V} \quad \text{!} \text{ — this would clip!}$$

Step 4 — Why 200kΩ Changes Things

With $R_f = 200 \text{ k}\Omega$ $R_f = 200 \text{ k}\Omega$:

$$Z_T = 20 \log_{10}(200 \times 103) = 106 \text{ dB} \quad Z_T = 20 \log_{10}(200 \times 103) = 106 \text{ dB}$$

$$V_{\text{TIA}}(0.5\text{m}) = 0.53 \text{ } \mu\text{A} \times 200 \text{ k}\Omega = 106 \text{ mV} \quad V_{\text{TIA}}(0.5\text{m}) = 0.53 \text{ } \mu\text{A} \times 200 \text{ k}\Omega = 106 \text{ mV}$$

$$V_{\text{TIA}}(0.1\text{m}) = 13 \text{ } \mu\text{A} \times 200 \text{ k}\Omega = 2.6 \text{ V} \quad V_{\text{TIA}}(0.1\text{m}) = 13 \text{ } \mu\text{A} \times 200 \text{ k}\Omega = 2.6 \text{ V}$$

After the bandpass filter with gain of 10:

$$V_{\text{filter}}(0.5\text{m}) = 106 \text{ mV} \times 10 = 1.06 \text{ V} \quad V_{\text{filter}}(0.5\text{m}) = 106 \text{ mV} \times 10 = 1.06 \text{ V} \quad \checkmark$$

$$V_{\text{filter}}(0.1\text{m}) = 2.6 \text{ V} \times 10 = 26 \text{ V} \quad V_{\text{filter}}(0.1\text{m}) = 2.6 \text{ V} \times 10 = 26 \text{ V} \quad \times \text{ Clips badly}$$

The Full Picture

	100kΩ TIA	200kΩ TIA
V_{out} at 0.5m (after filter)	530 mV	1.06 V
V_{out} at 0.1m (after filter)	Clips	Clips worse
TIA gain	100 dB	106 dB
Total system gain	120 dB	126 dB
Risk of clipping	Lower $\checkmark$	Higher !

Bottom Line

- Your photodiode produces **0.5 – 13 μA** depending on distance
- Your TIA at 100kΩ converts that to **53mV – 1.3V** — safe, no clipping $\checkmark$
- Your bandpass filter then amplifies and filters that signal
- Going to 200kΩ gives you **2x voltage** at range but **risks clipping up close**
- 100kΩ is the safer, more robust choice** for a system that needs to work at varying distances

Comparator

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Inverting Schmitt Trigger Comparator

Search this on Google Images and you'll see exactly the topology.

The Starting Point — What We Know

From your signal chain:

Parameter	Value	Where From
Noise floor	0.2 mV0.2 mV	$I_{dark} \times R_f = 2nA \times 100k \Omega I_{dark} \times R_f = 2nA \times 100k \Omega$
Weakest real signal	13 mV13 mV	$IPD(1m) \times R_f = 0.13\mu A \times 100k \Omega IPD(1m) \times R_f = 0.13\mu A \times 100k \Omega$
Supply voltage	3.3 V3.3 V	Your system supply
Output HIGH	3.3 V3.3 V	Pull-up to 3.3V
Output LOW	0 V0 V	LM393 open collector pulls to GND

The Design Requirements

You need:

$V_{TL} > V_{noise} = 0.2 \text{ mV}$ $V_{TL} > V_{noise} = 0.2 \text{ mV}$

$V_{TH} < V_{signal, min} = 13 \text{ mV}$ $V_{TH} < V_{signal, min} = 13 \text{ mV}$

$V_{hysteresis} = V_{TH} - V_{TL} > 0$ (wide enough to prevent chatter)

$V_{hysteresis} = V_{TH} - V_{TL} > 0$ (wide enough to prevent chatter)

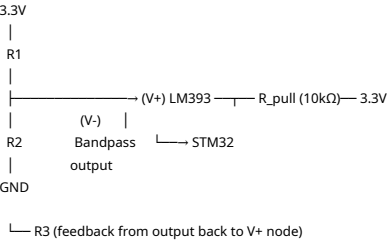
Choosing sensible values within these constraints:

$V_{TH} = 8 \text{ mV}$ $V_{TH} = 8 \text{ mV}$

$V_{TL} = 3 \text{ mV}$ $V_{TL} = 3 \text{ mV}$

$V_{hysteresis} = V_{TH} - V_{TL} = 8 - 3 = 5 \text{ mV}$ $V_{hysteresis} = V_{TH} - V_{TL} = 8 - 3 = 5 \text{ mV}$

The Circuit — Non-Inverting Schmitt Trigger



The thresholds are set by **three resistors**:

- R1R1 and R2R2 — voltage divider setting the **centre threshold**
- R3R3 — feedback resistor controlling the **hysteresis width**

Step 1 — Find The Centre Threshold

The centre threshold is the midpoint between V_{TH} and V_{TL} :

$V_{centre} = \frac{V_{TH} + V_{TL}}{2} = \frac{8 + 3}{2} = 5.5 \text{ mV}$ $V_{centre} = \frac{V_{TH} + V_{TL}}{2} = \frac{8 + 3}{2} = 5.5 \text{ mV}$

This is set by the R1R1, R2R2 voltage divider:

$V_{centre} = V_{supply} \times \frac{R2}{R1 + R2}$ $V_{centre} = V_{supply} \times \frac{R2}{R1 + R2}$

$R2 = \frac{V_{centre} \times (R1 + R2)}{V_{supply} - V_{centre}}$ $R2 = \frac{5.5 \text{ mV} \times (1k\Omega + R2)}{3.3V - 5.5 \text{ mV}}$

Choose R2 = 1 kΩ, then:

$R1 = \frac{R2 \times V_{supply}}{V_{supply} - V_{centre}} - R2$ $R1 = \frac{1k\Omega \times 3.3V}{3.3V - 5.5 \text{ mV}} - 1k\Omega \approx 598k\Omega$

Step 2 — Find The Hysteresis Resistor R3

The hysteresis band is controlled by how much the output feeds back to shift the threshold. The formula is:

$V_{hysteresis} = V_{supply} \times \frac{R3}{R1 + R3}$ $V_{hysteresis} = V_{supply} \times \frac{R3}{R1 + R3}$

First find R1 || R2:

$R1 || R2 = \frac{R1 \times R2}{R1 + R2} = \frac{598k\Omega \times 1k\Omega}{598k\Omega + 1k\Omega} \approx 998\Omega$ $R1 || R2 = \frac{598k\Omega \times 1k\Omega}{598k\Omega + 1k\Omega} \approx 998\Omega$

Now rearrange for R3:

$R3 = \frac{V_{hysteresis} \times (R1 || R2)}{V_{supply} - V_{hysteresis}}$ $R3 = \frac{5 \text{ mV} \times 998\Omega}{3.3V - 5 \text{ mV}}$

$R3 = \frac{5 \text{ mV} \times 998\Omega}{3.3V - 5 \text{ mV}}$ $R3 = \frac{5 \text{ mV} \times 998\Omega}{3.3V - 5 \text{ mV}}$

$R3 = \frac{5 \text{ mV} \times 998\Omega}{3.3V - 5 \text{ mV}}$ $R3 = \frac{5 \text{ mV} \times 998\Omega}{3.3V - 5 \text{ mV}}$

Step 3 — Verify The Thresholds

With $R1 = 560 \text{ k}\Omega$, $R2 = 1 \text{ k}\Omega$, $R3 = 680 \text{ k}\Omega$, $R4 = 680 \text{ k}\Omega$:

When output is **HIGH** (3.3V), $R3$ pulls the threshold **up**:

$$V_{TH} = V_{supply} \times \frac{R2}{R1 + R2} \parallel R3 + R2 = V_{supply} \times \frac{R2}{R1 + R2 + R3}$$

$$R1 \parallel R3 = \frac{560 \text{ k} \times 680 \text{ k}}{560 \text{ k} + 680 \text{ k}} = 307.8 \text{ k}\Omega$$

$$R1 \parallel R3 = \frac{560 \text{ k} \times 680 \text{ k}}{560 \text{ k} + 680 \text{ k}} = 307.8 \text{ k}\Omega$$

$$V_{TH} = 3.3 \times \frac{1000}{307800 + 1000} = 3.3 \times 0.00325 = 10.7 \text{ mV}$$

$$V_{TH} = 3.3 \times \frac{1000}{307800 + 1000} = 3.3 \times 0.00325 = 10.7 \text{ mV}$$

When output is **LOW** (0V), $R3$ pulls the threshold **down**:

$$V_{TL} = V_{supply} \times \frac{R2}{R1 + R2 + R3} \parallel R4$$

$$V_{TL} = 3.3 \times \frac{1000}{560000 + 1000 + 680000} = 5.5 \text{ mV}$$

$$V_{hysteresis} = V_{TH} - V_{TL} = 10.7 - 5.5 = 5.2 \text{ mV}$$

Final Component Values

Component	Value	Purpose
$R1$	560 k Ω	Top of threshold divider
$R2$	1 k Ω	Bottom of threshold divider
$R3$	680 k Ω	Hysteresis feedback
$R4$	10 k Ω	Pull-up on open collector output

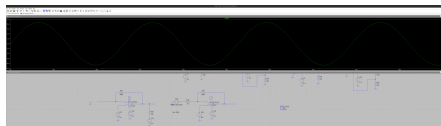
Summary Of What Each Value Achieves

$V_{TH} \approx 10.7 \text{ mV}$ — **below** your 13mV minimum signal ✓

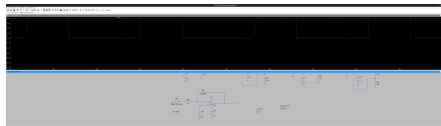
$V_{TL} \approx 5.5 \text{ mV}$ — **above** your 0.2mV noise floor ✓

$V_{hysteresis} \approx 5 \text{ mV}$ — wide enough to prevent chatter ✓

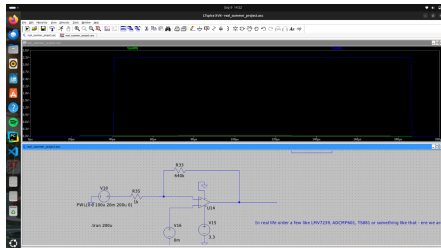
The circuit will cleanly switch HIGH when your target is detected and stay LOW through all noise — exactly what your STM32 needs.



Linear active region



Working but a delay



Trainlur waveform. Works perfectly

Testing overall circuit

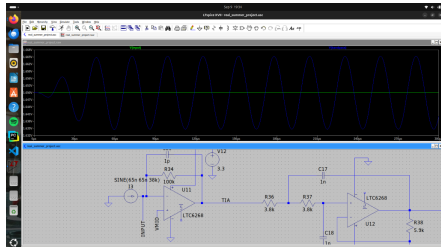
Wednesday, September 09, 2026 3:06 PM

Running .op with input signal with 0 amplitude to test DC bias we find this:
V(bandpass): 0.000673484 voltage

With **zero input signal** (no target present), your comparator sees **0.673 mV** sitting at its input — not 1.65V, not some random DC offset. It's sitting right near ground, comfortably **below your 3 mV low threshold**.

This is exactly what you want. It means:

- ✓ No DC bias mismatch — your filter output is correctly centered near 0V, not stuck at some half-supply voltage
- ✓ With no target, your comparator input sits below $V_{TL}/V_{TL} = 3\text{mV}$, so the comparator will correctly report "no target" and stay LOW



Input amplitude = 65n

After bandpass amplitude = 0.015

$0.015/65\text{n} = 2.24 \times 10^5 = 107\text{dB}$ which is the gain we wanted, as shown in ac sweep. This means that the issue was the clipping and the circuit now works.

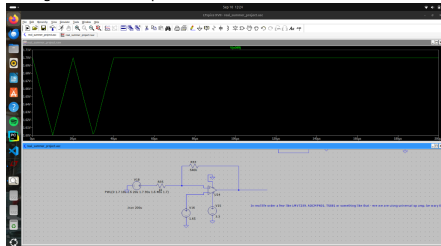
Real photodiode amplitude values maximum is

$1.65/2.24 \times 10^5 = 7.4\mu$

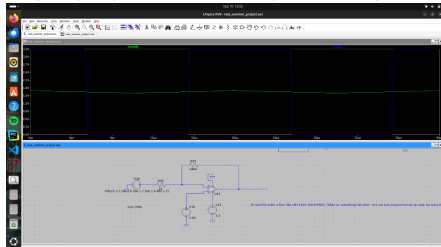
The analog front-end was biased at mid-rail and the TIA gain was selected to balance sensitivity and headroom. In simulation, the circuit remained linear up to about $7.4\mu\text{A}$ input photocurrent, indicating that weak signals can still be amplified effectively while strong reflections do not clip prematurely.

Now must bias the comparator to **1.653 V** and **1.647 V**

Testing the biased comparator

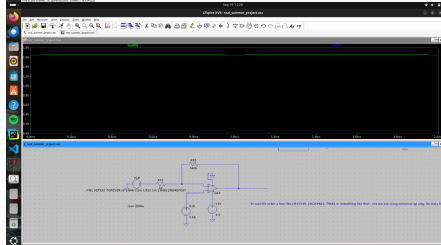
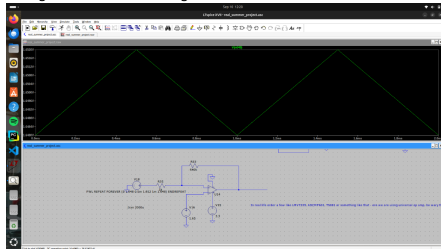


Using an input square wave we see that input should be high until about 5u, then low till



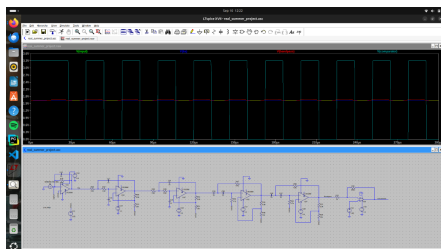
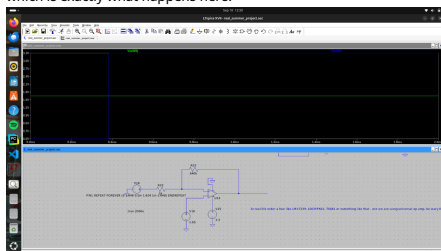
Works

Now gonna test with a triangular waveform in the bounds



Worjs

Doing a test where a traingular waveform that goes between 1.654 V and 1.648 V, so it shold go high then just shy of the lower bund of the hysteresis therefore reamin high, which is exactly what happens here.



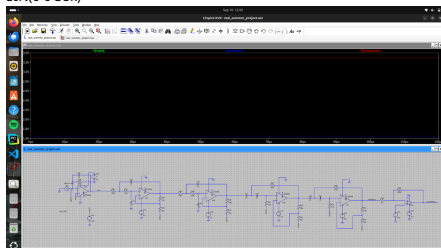
Works

The MCU will read the sqaure wavea nd internpret it as hey theres a target.

Final test cases

No target / dark case

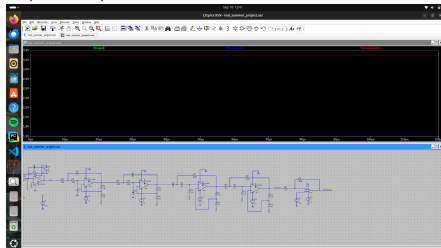
SIN(0 0 38k)



Should be constantly low, not constantly high.

No target but ambient light present

SIN(500n 0 38k)

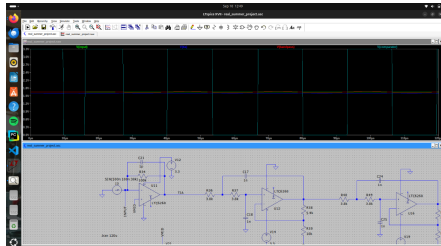


Same issue here

Instead of tinkering with the crcuit, it would be easier to just say if the ouputs constant high, then assume no target (the MCU processes this part)

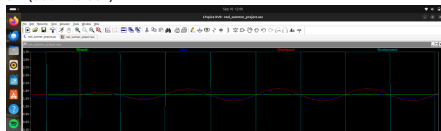
Normal target

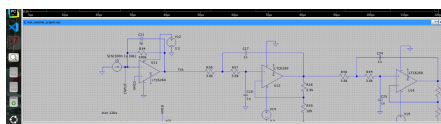
SIN(500n 100n 38k)



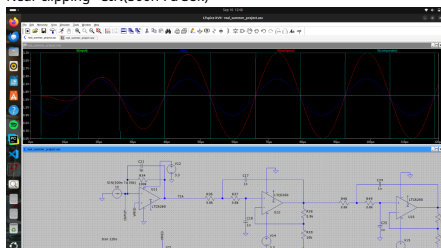
Strong target

SIN(500n 1u 38k)



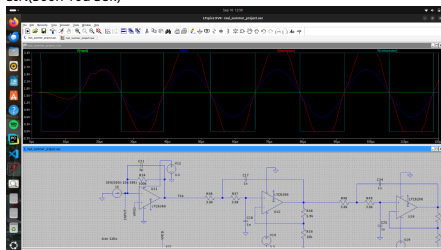


Near clipping - SIN(500n 7u 38k)



Clipping

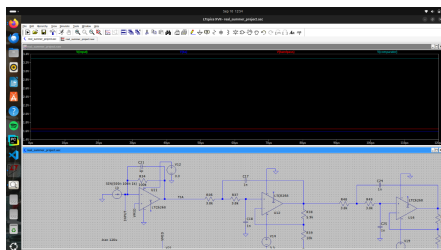
SIN(500n 10u 38k)



Everything works – just need to bare in mind for the MCU that constant high or low means ignore it.

Wrong low frequency

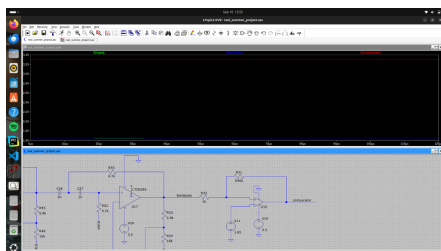
SIN(500n 100n 1k)



Constant high input

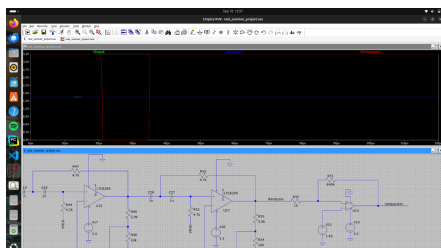
Wrong mid-low frequency

SIN(500n 100n 10k)



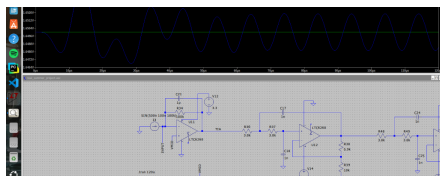
Wrong high frequency

SIN(500n 100n 100k)



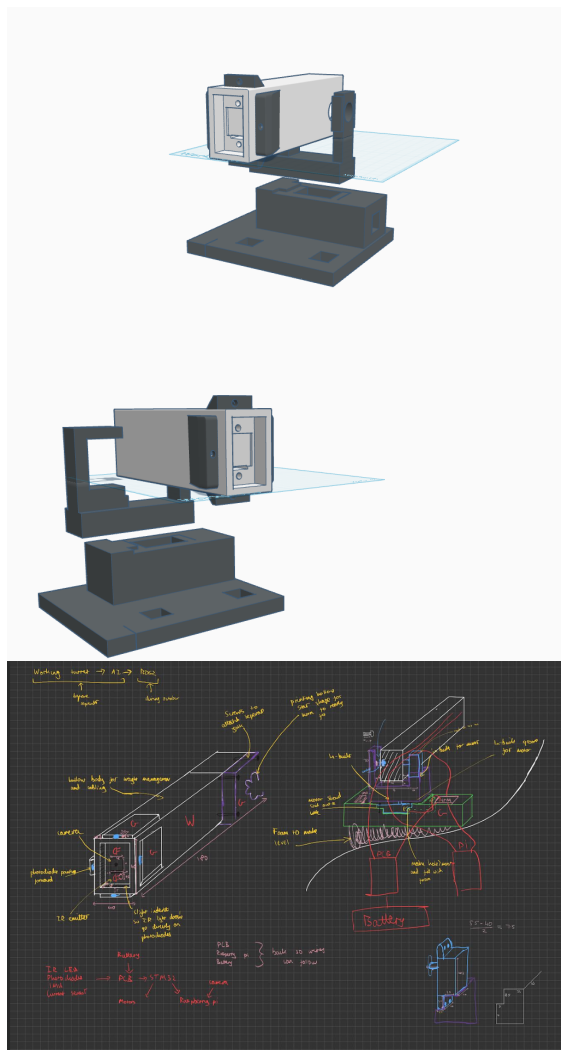
When I look closely at the bandpass signal I see this





Due to transients or whatnot theres some complex stuff going on but since its not a constant square wave it doesn't matter. I could increase the hysteresis band but its fine.

Friday, August 28, 2026 8:47 PM



Less justifications and time spent here as I am focused on the electronics/software side, not mechanical side.

